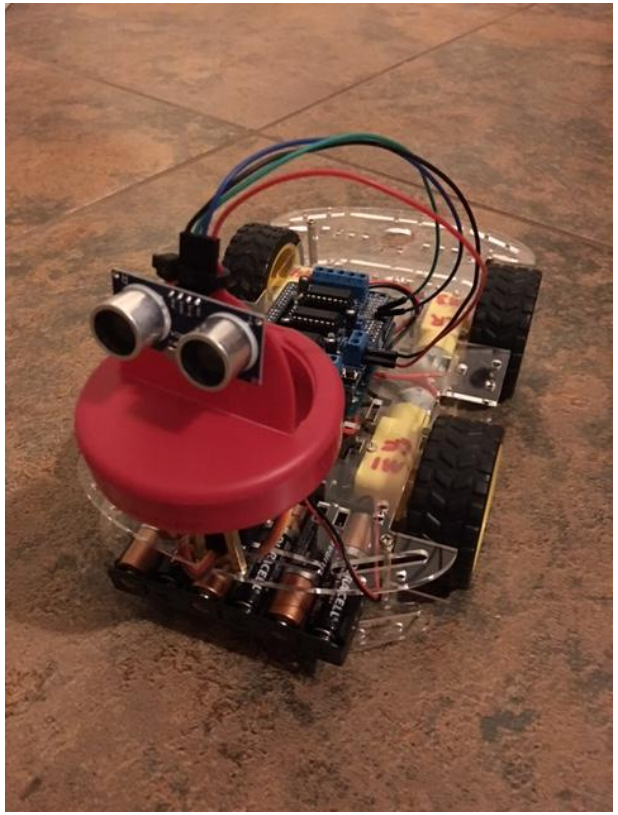
ENGR 1201 – Introduction to Engineering

Austin Community College

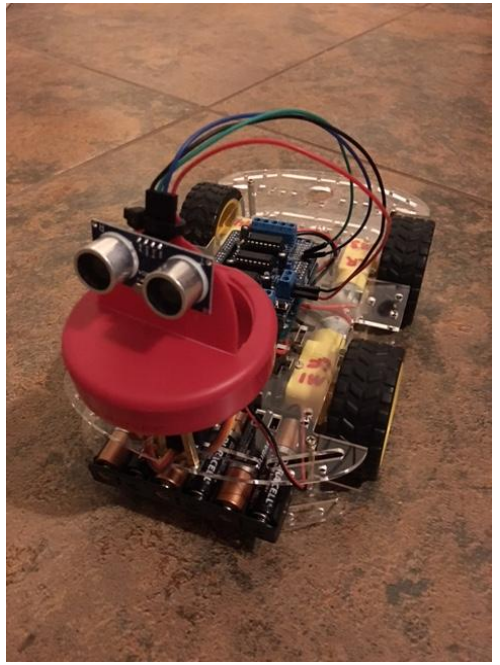
ENGR1201 Group Project: Riverbot Car



- Design and build an obstacle-avoiding car
  - Car must move
  - Car must avoid obstacles
  - Customers prefer:
    - Fast-moving cars
    - Cars that don’t break if they accidentally encounter obstacle
    - Cars with easy access to the on/off switch
    - Cars whose batteries are easy to remove for charging
    - Cars that travel in straight lines in absence of obstacles
    - Extra features that differentiate the car
- 3-4 students per group

ENGR1201 Group Project: Riverbot Car

- Competitive analysis info from previous ACC and A&M classes:
  - Prof. Shana Shaw discusses group project: https://www.youtube.com/watch?v=DWe6WIP3cbw
  - Spring 2022 semester: https://youtu.be/GxlHM_gV2GE
  - Fall 2022 semester: https://www.youtube.com/watch?v=wg2zhO0YrOY
  - Students' cars from Spring 2023 semester: https://www.youtube.com/watch?v=w8gk2eVenRM



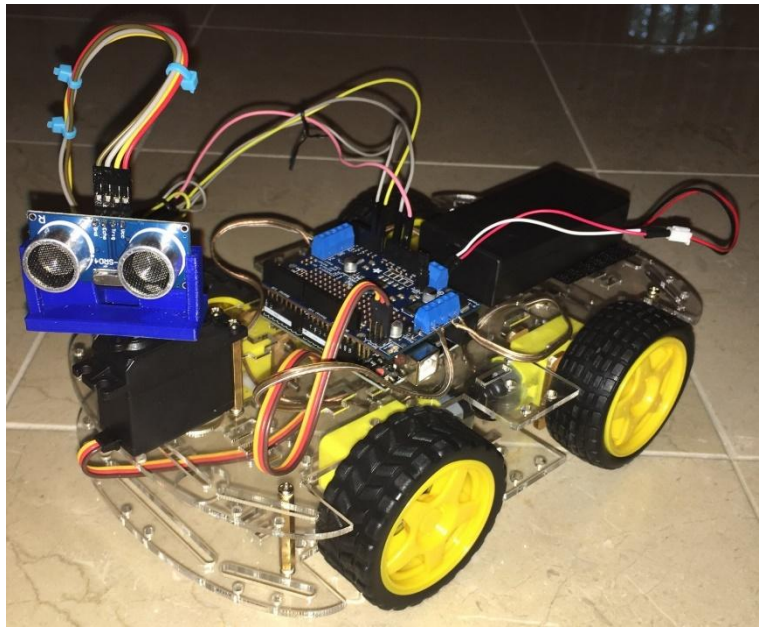
Your Team's Robot Kit

- Your team may take DeWalt box, returning it on/by last day of class.



- You will need:
  - Arduino board (1)
  - USB cable (1)
  - Motor-shield board (1)
  - Ultrasonic sensor
  - Small or large servo motor (1)
  - Breadboard
  - DC motors (4) with mounts, screws
  - Wheels (4)
  - Acrylic chassis plates (2)
  - Connection wires
  - Screws, nuts, spacers
  - Misc. other parts
- Other parts available upon request, subject to availability

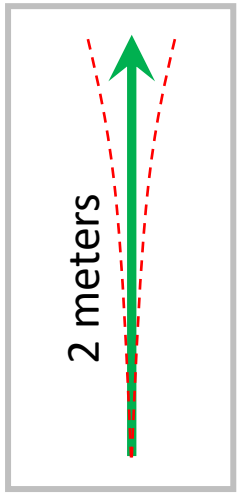
- ***Kit lacks a laptop***; someone from your team should install Arduino IDE software on their computer for use outside of class.



Two Assignments Due in One Week

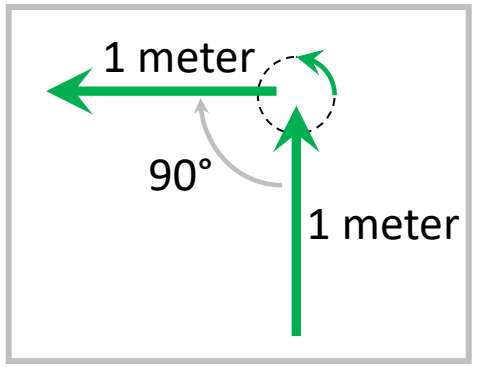
A. 1) Assemble your car and create a sketch that allows the car to travel in a straight line

- Car must travel 2.00 meters without deviating from a straight line by more than 0.10 m



2) Create a sketch that allows the car to:

a. Travel straight for 1.0±0.1 m, and then
b. Stop and turn left by 90±10°, and then
c. Travel straight for 1.0±0.1 m, and then
d. Stop for 5 seconds

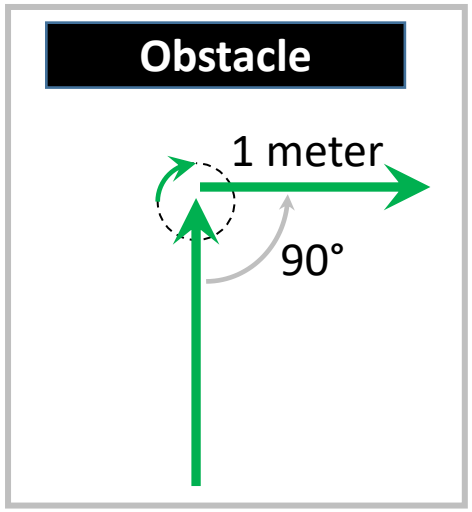


Two Assignments Due in Two Weeks

B. Create a sketch that allows the car to:

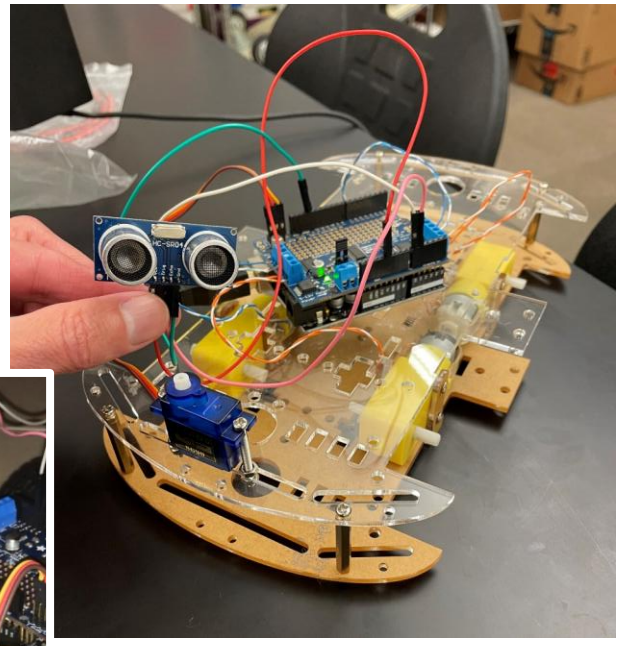
a. Travel straight towards an obstacle,
b. Detect the obstacle with ultrasonic sensor and stop, then
c. Turn right by 90±10°, and then
d. Travel straight for 1.0±0.1 m, and then
e. Stop for 5 seconds.

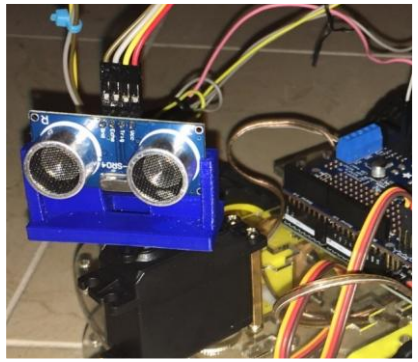
- Hint #1: This requires an ultrasonic sensor.
- Hint #2: The sensor can be taped to front of car without a servo motor.



C. Use CAD software to design a bracket to mount your ultrasonic sensor to your car's servo motor

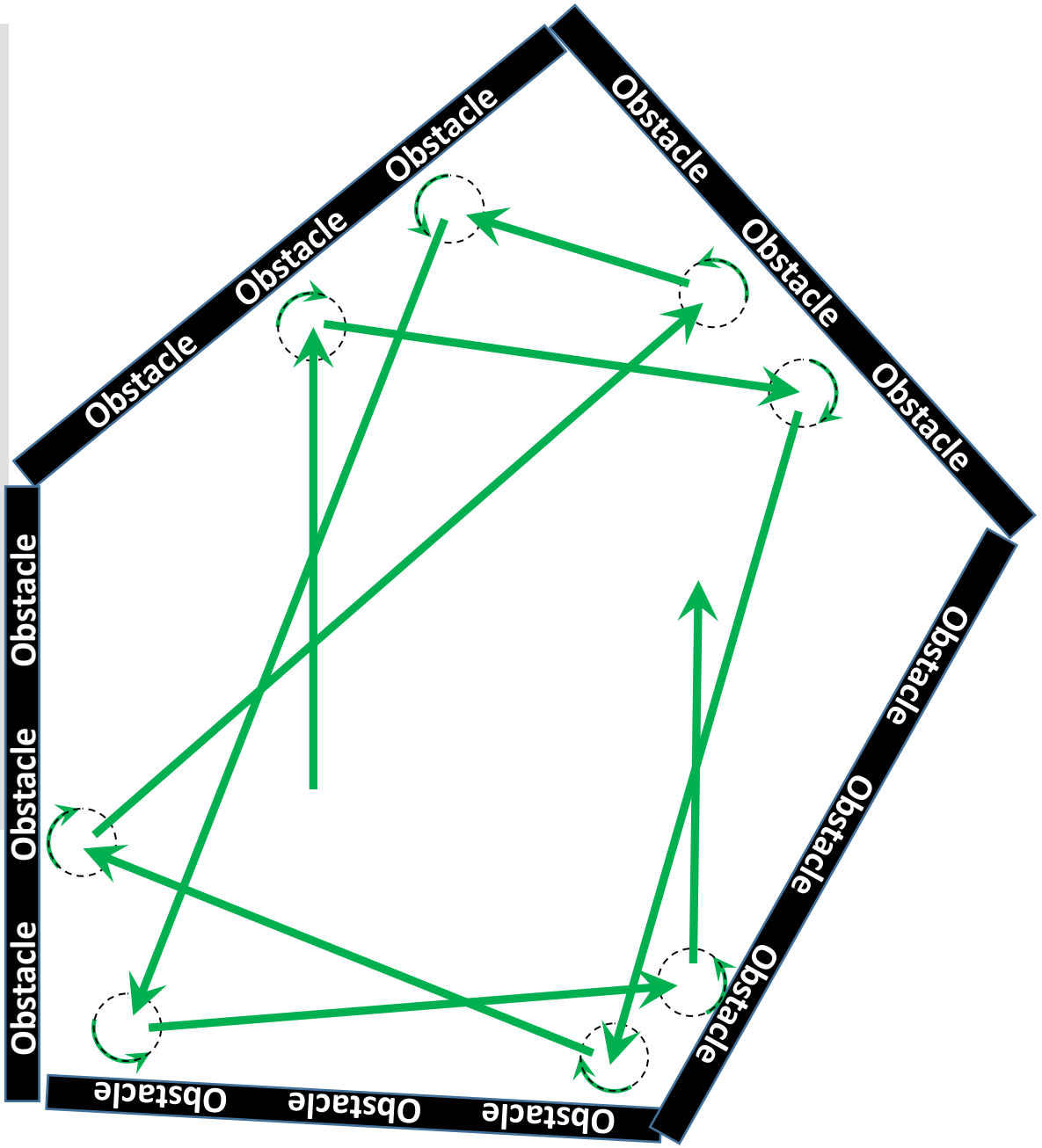
- Use digital calipers to measure your ultrasonic sensor
- You may 'reverse-engineer' the bracket on ACC's car



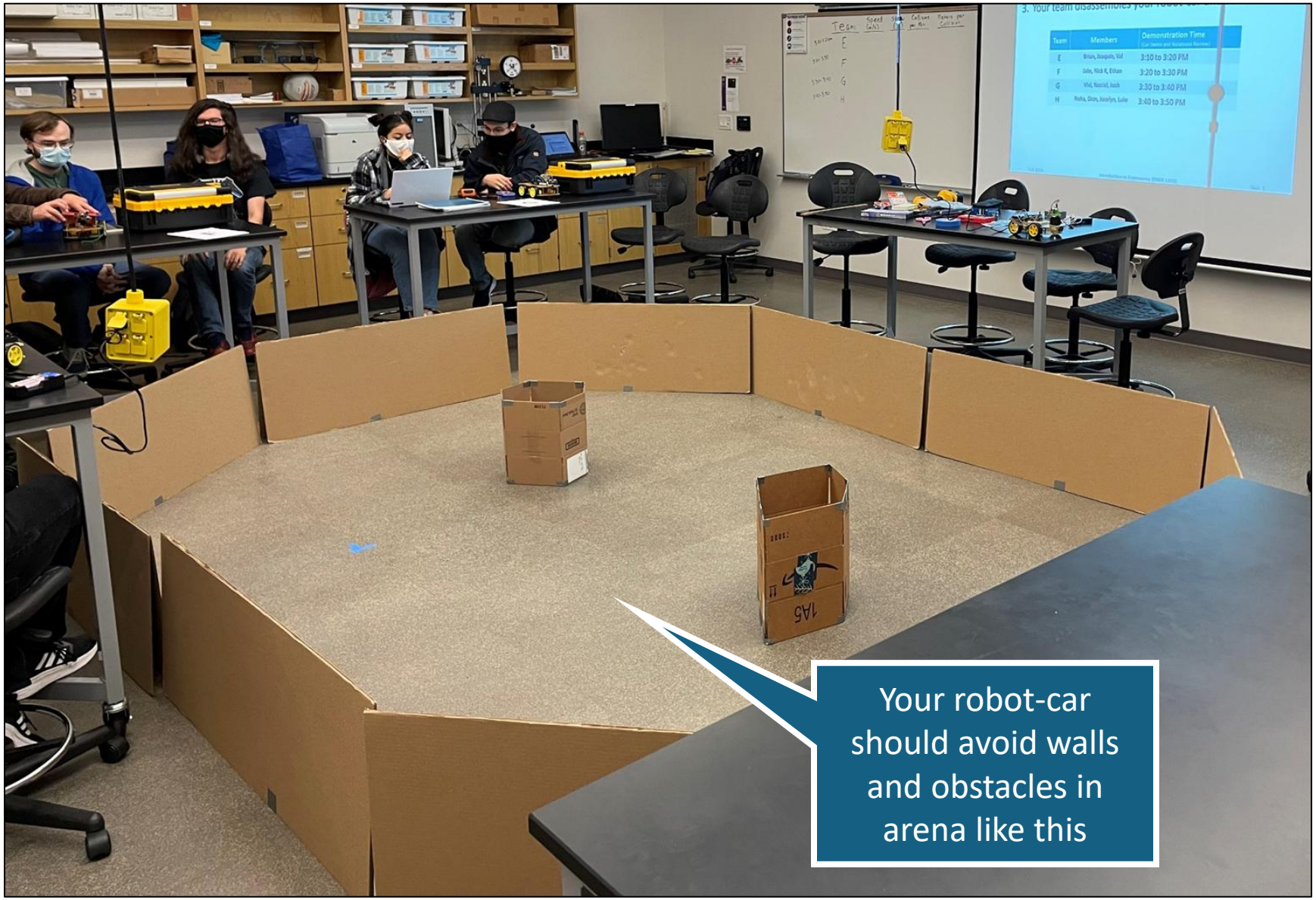


Due in Three Weeks

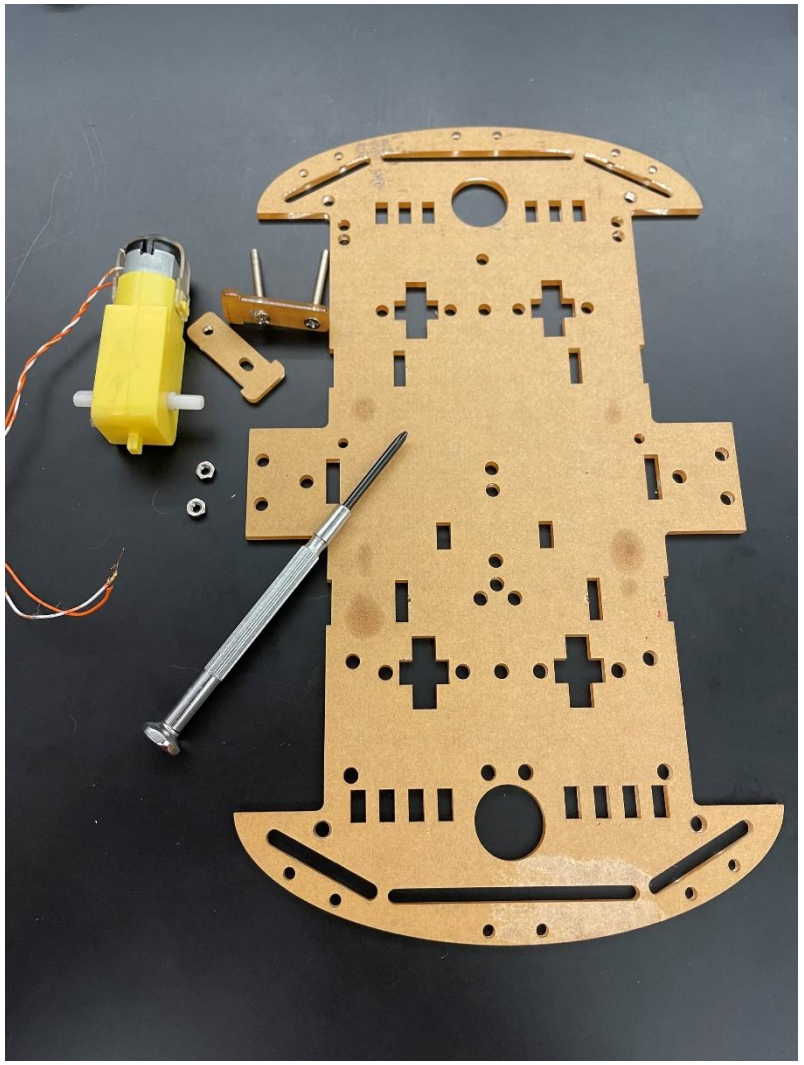
- Demonstrate preliminary version of obstacle-avoiding robot-car.
- Car must use ultrasonic sensor and a servo motor to avoid obstacles
- You may use ACC's C++ sketch to get started
- Upload your final C++ program to Blackboard.

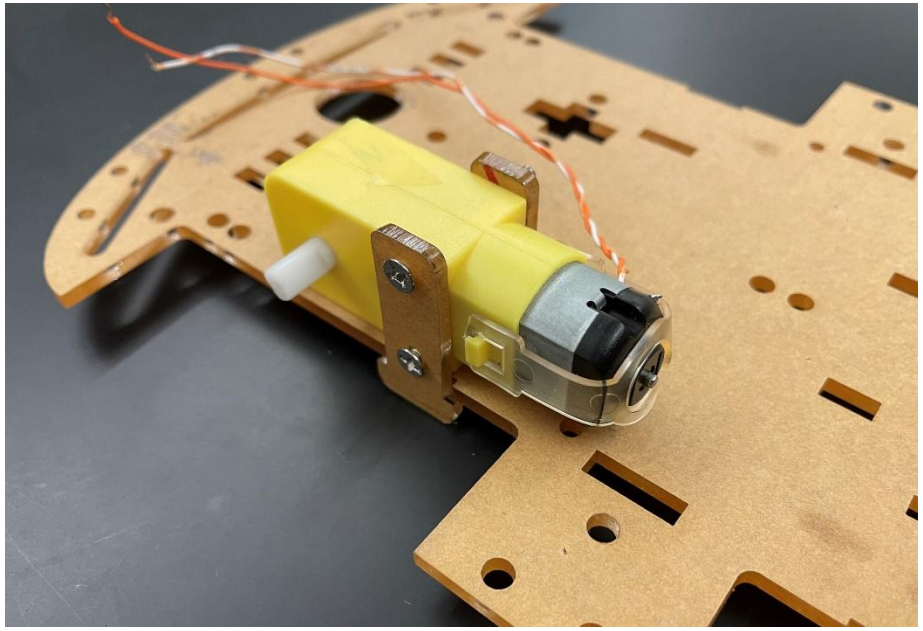


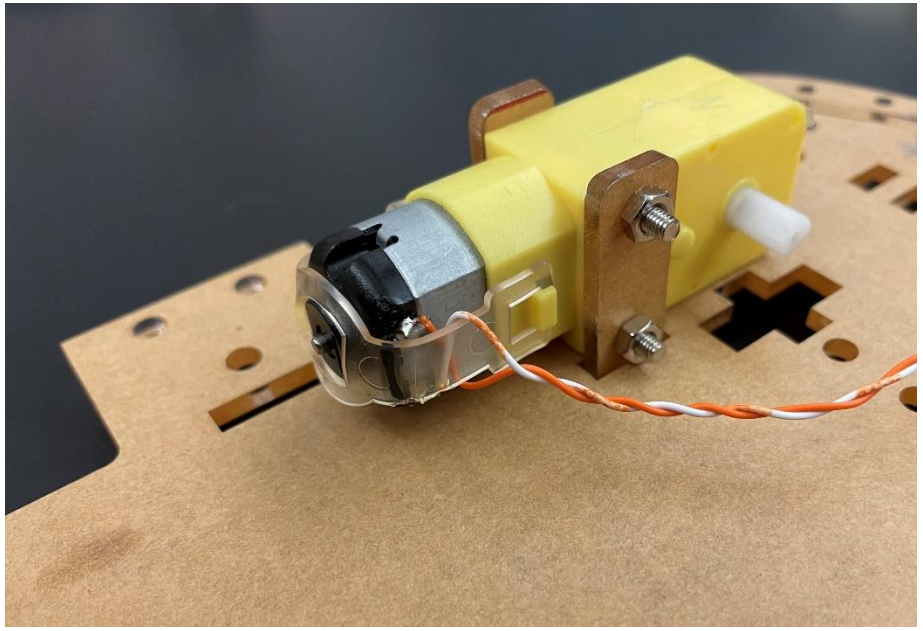
Object-Avoidance Testing



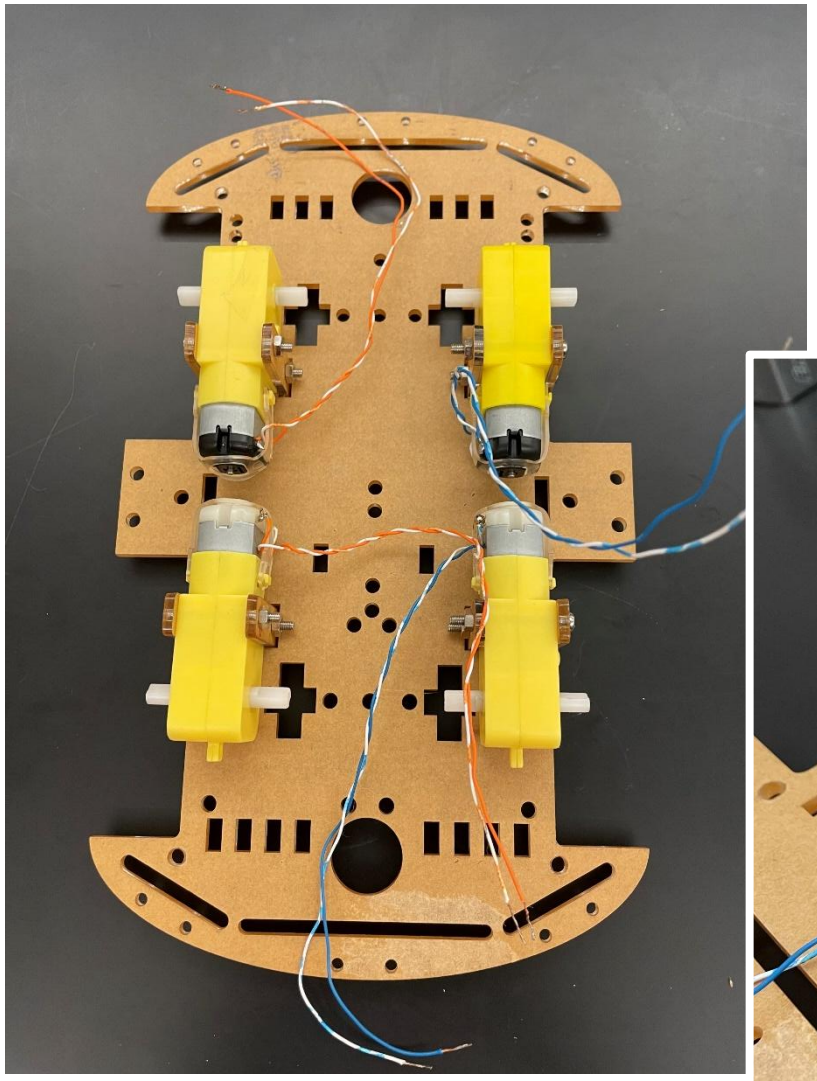
Assembling a Robot Car

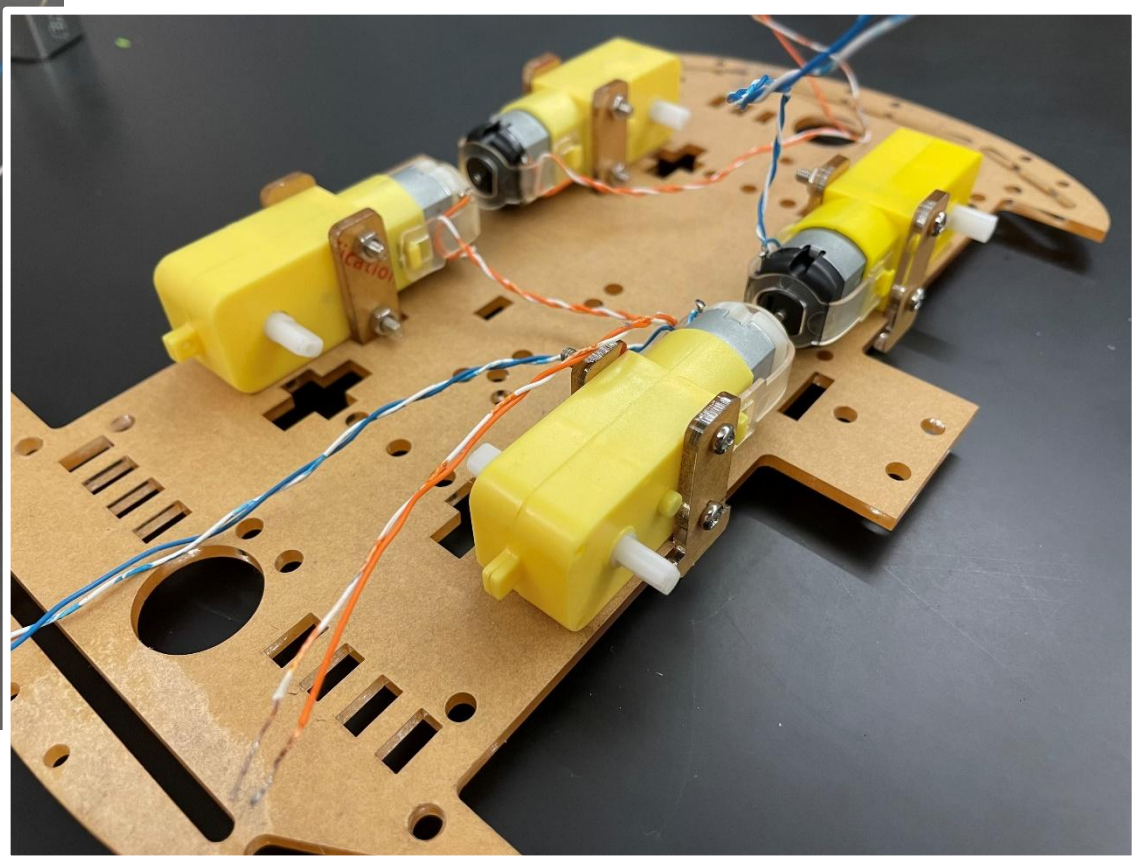




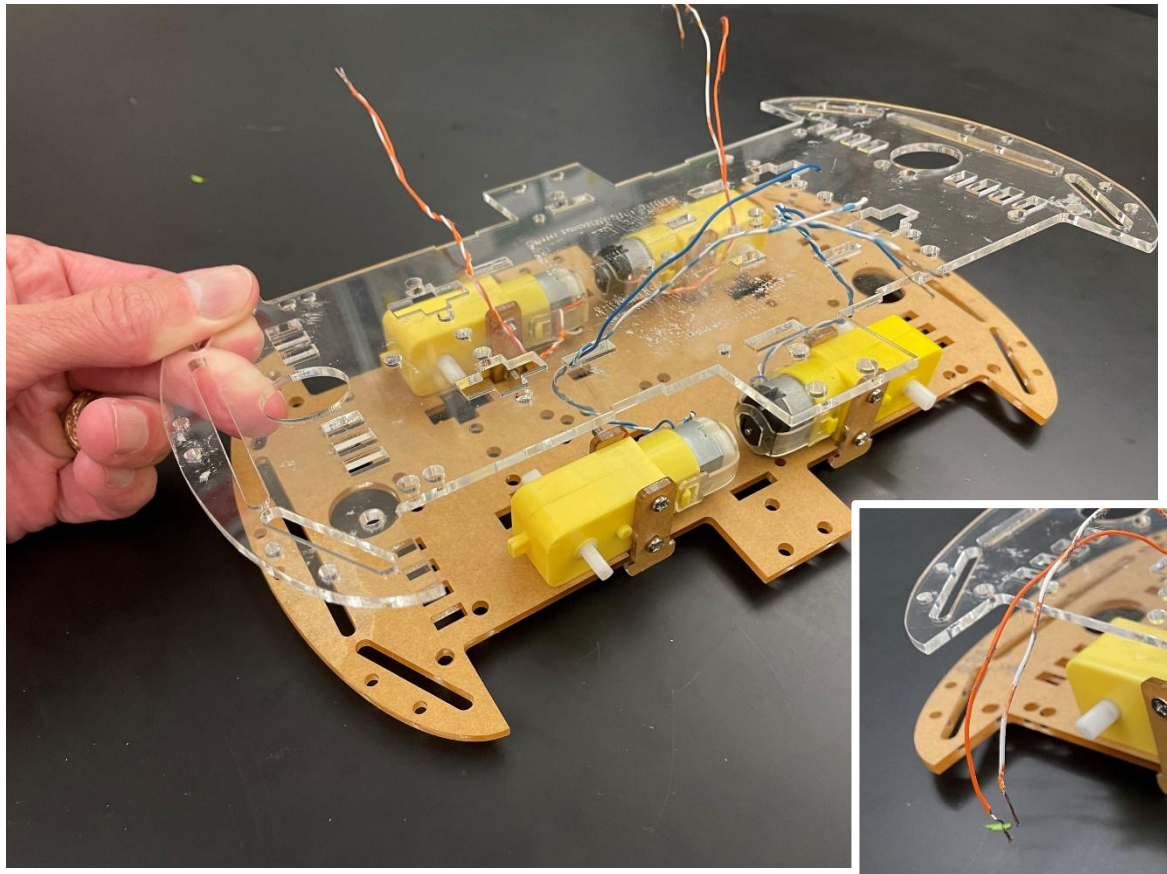


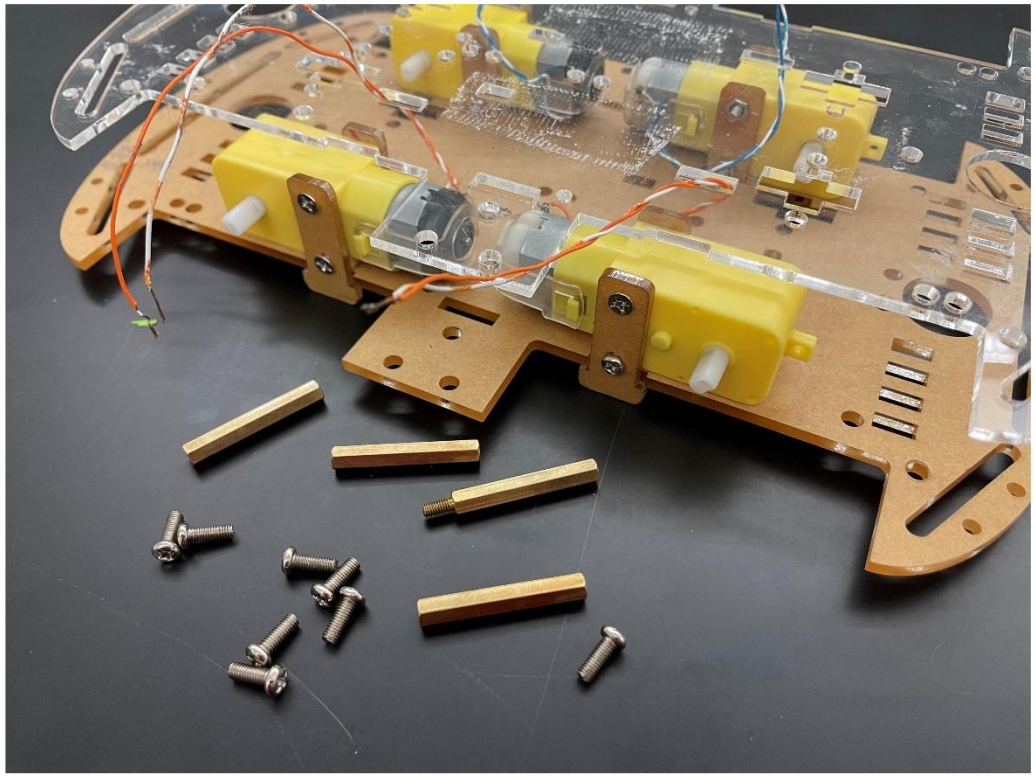
Assembling a Robot Car (cont.)



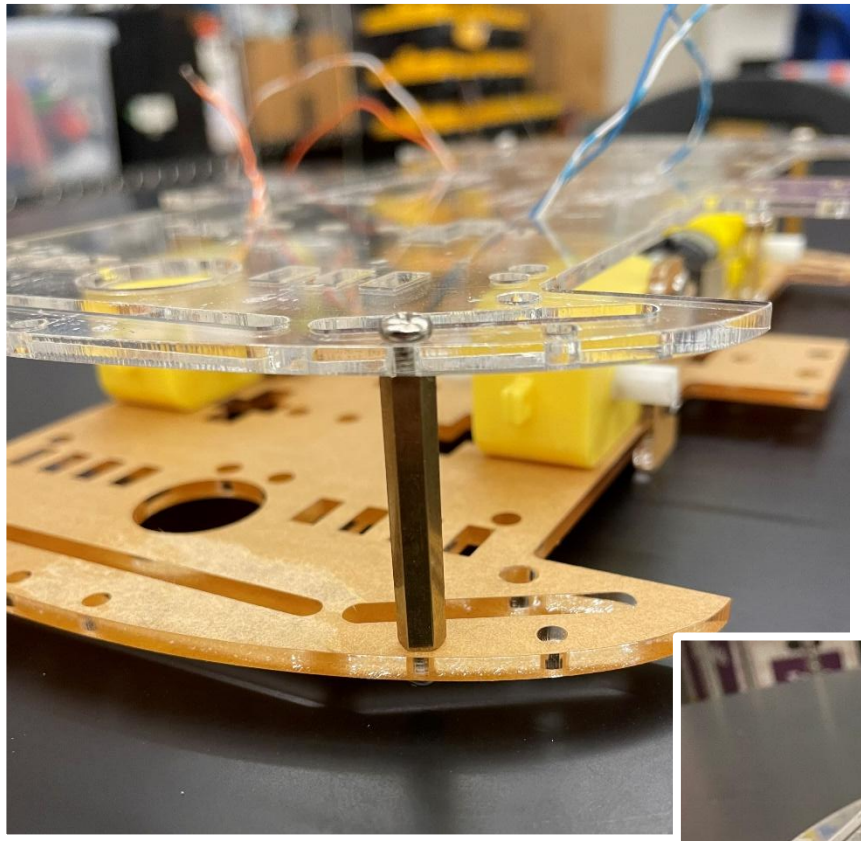


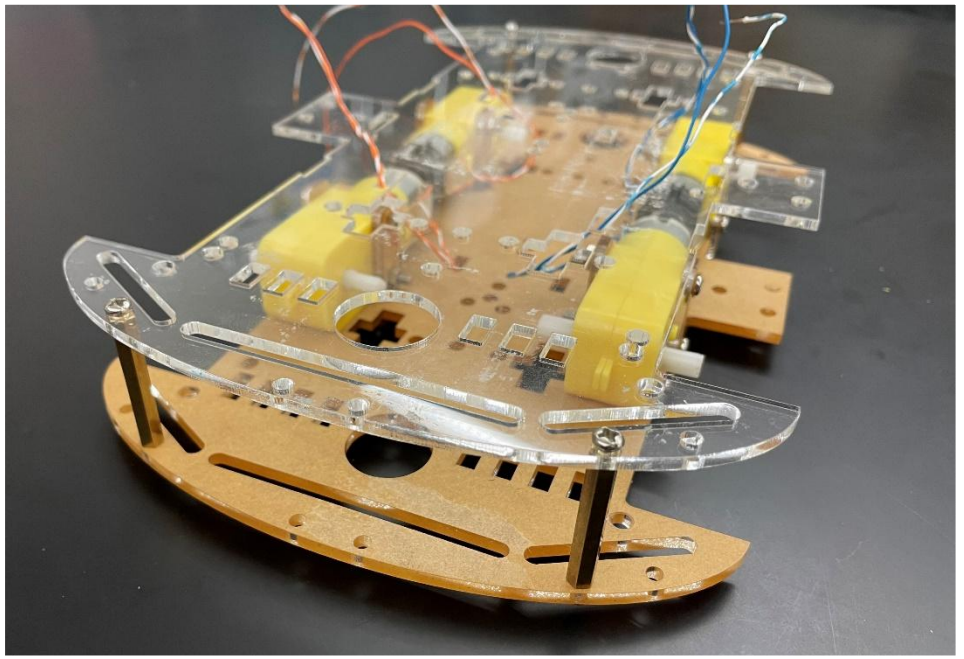
Assembling a Robot Car (cont.)

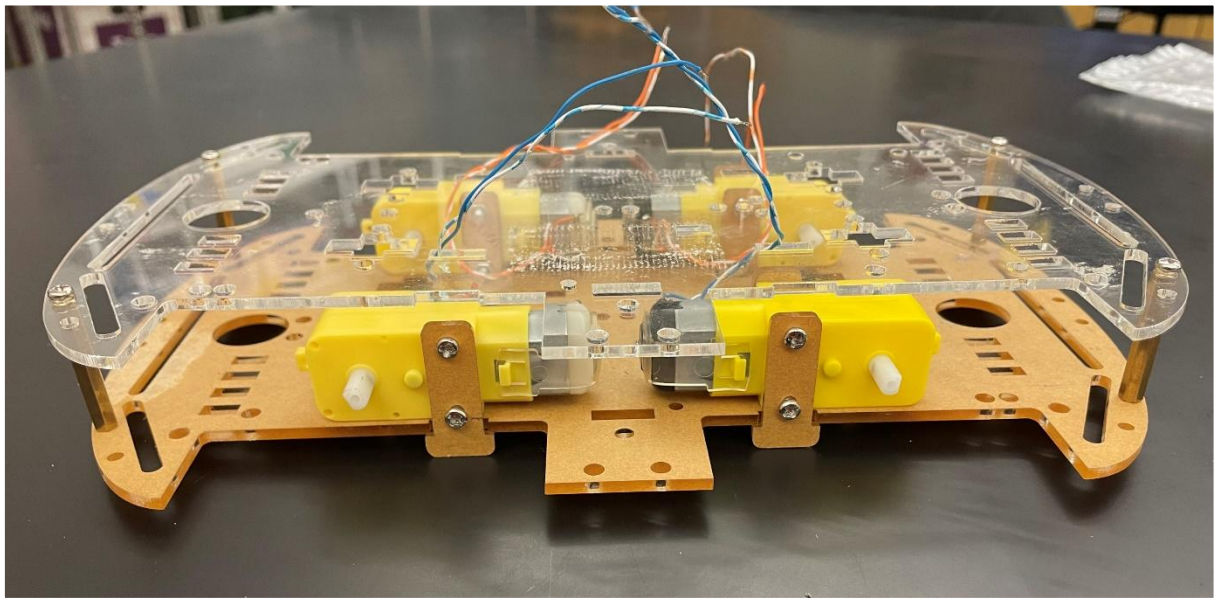




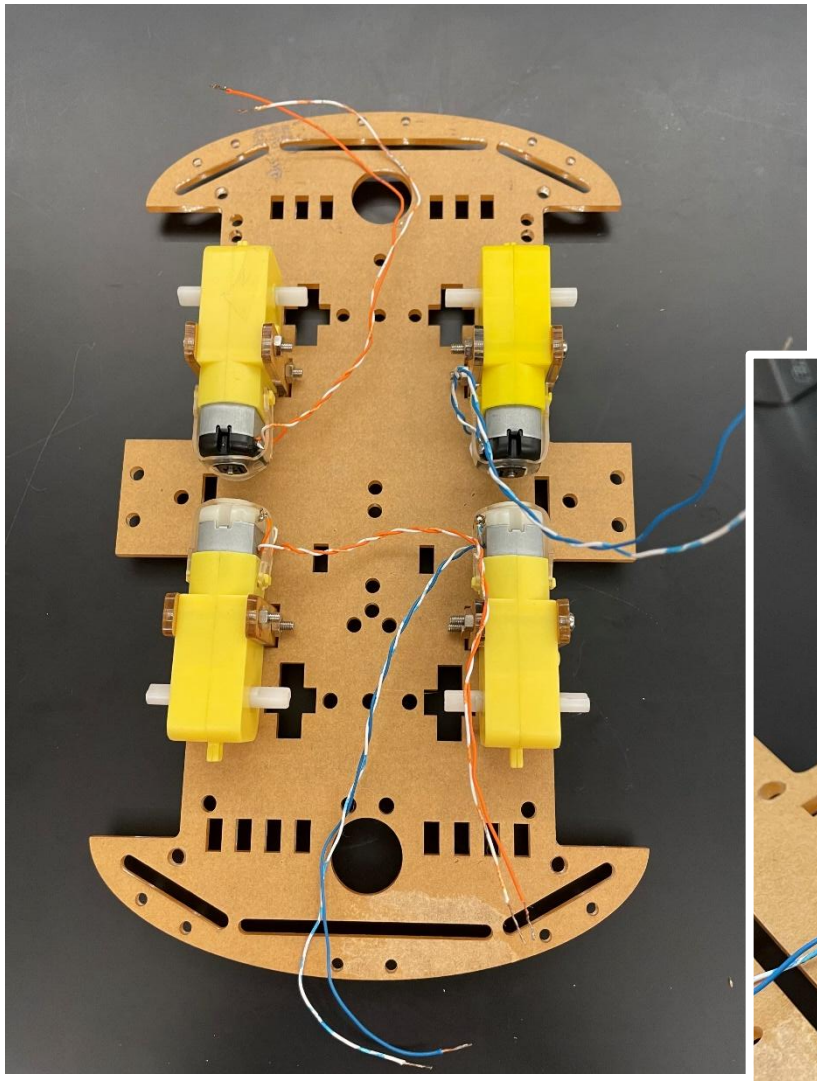
Assembling a Robot Car (cont.)

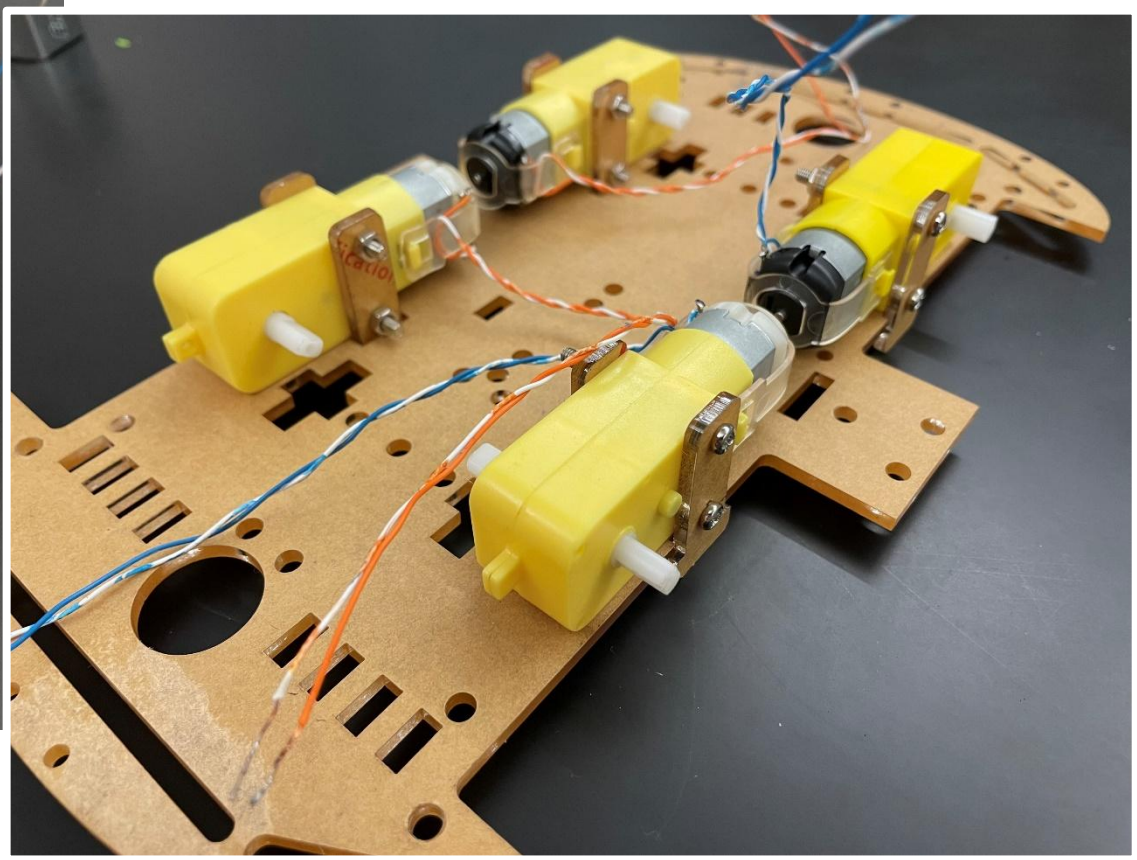




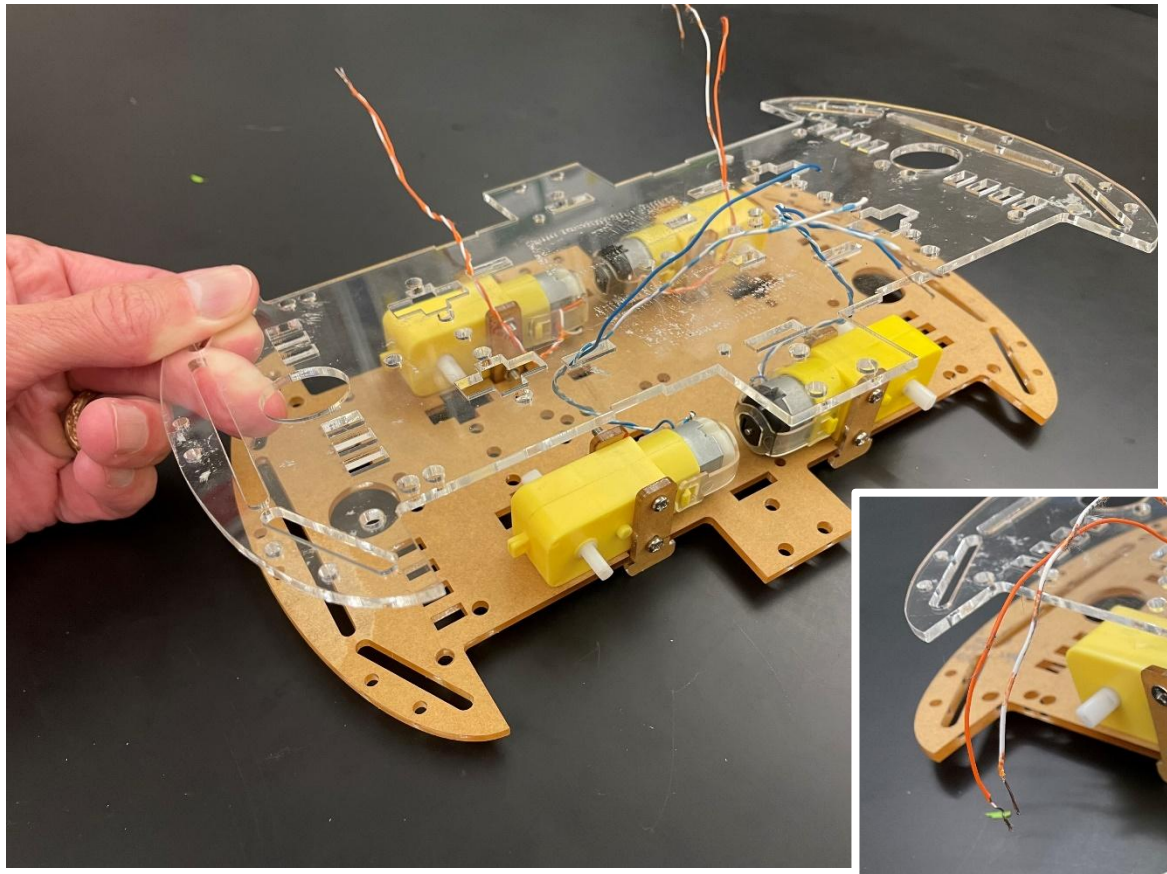


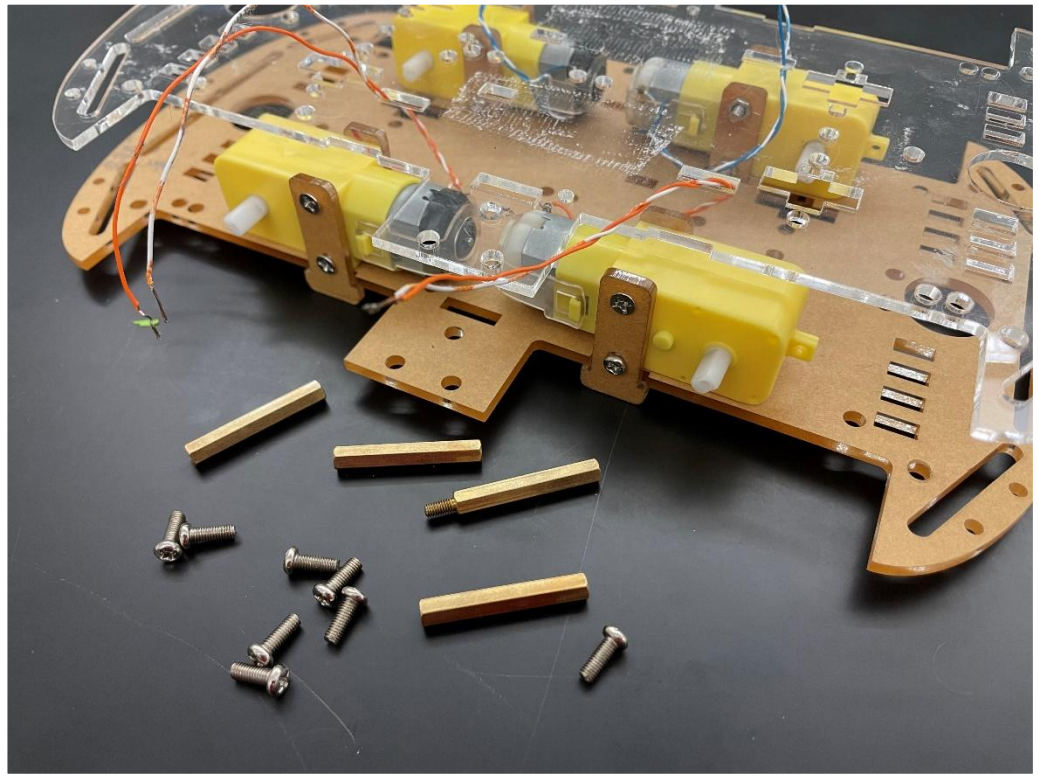
Assembling a Robot Car (cont.)



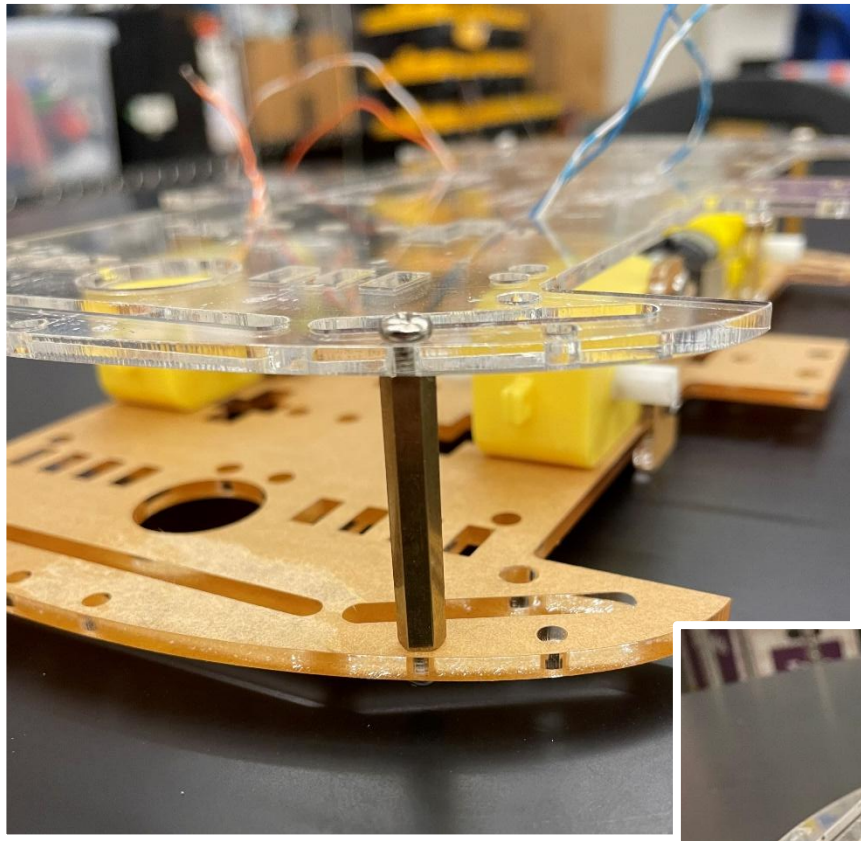


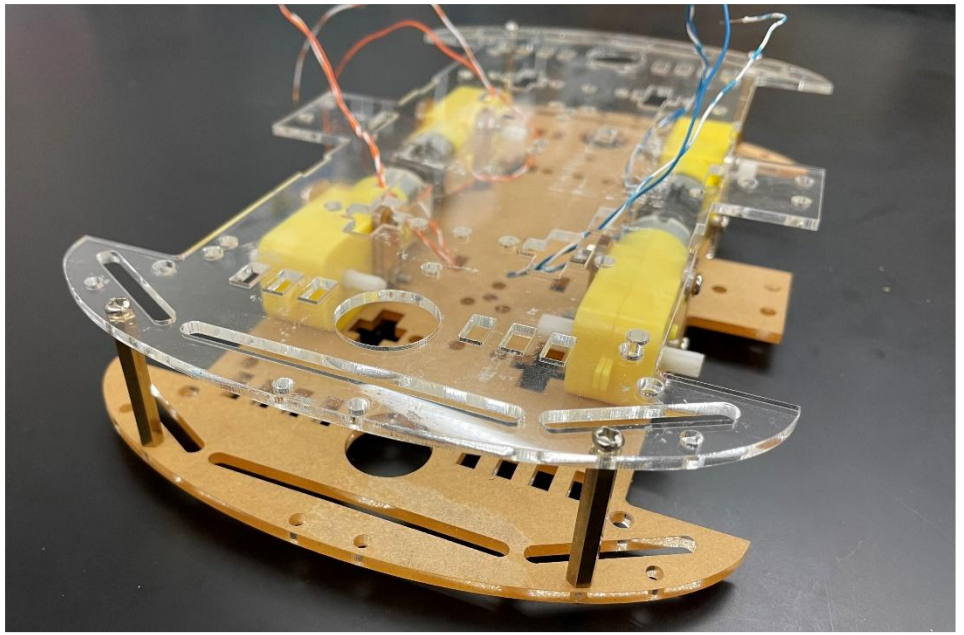
Assembling a Robot Car (cont.)

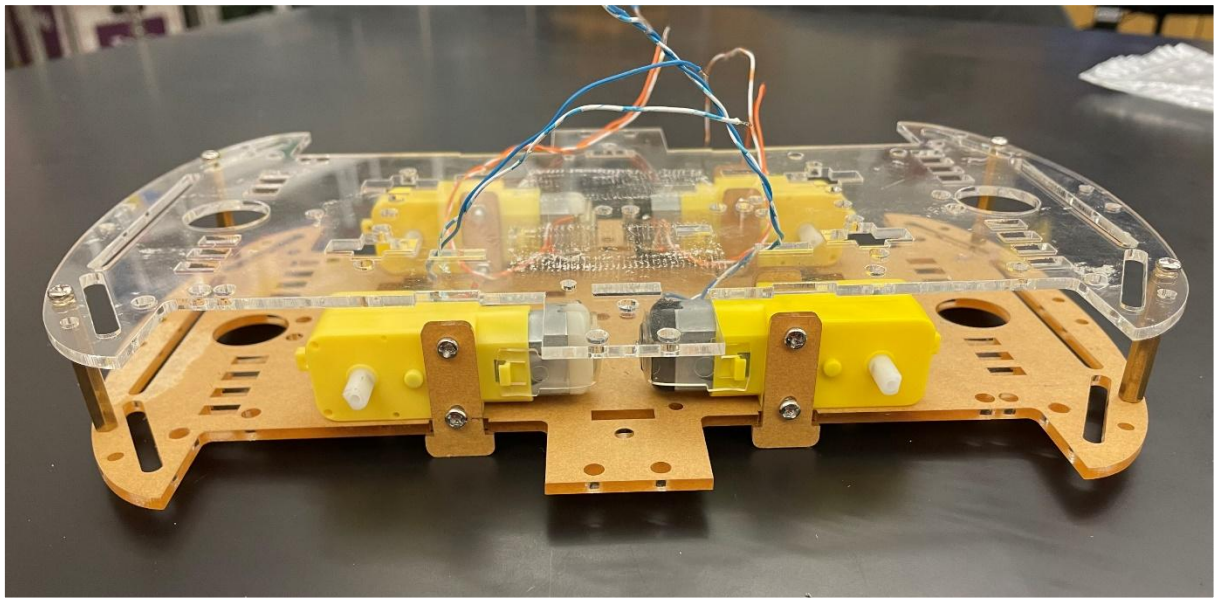




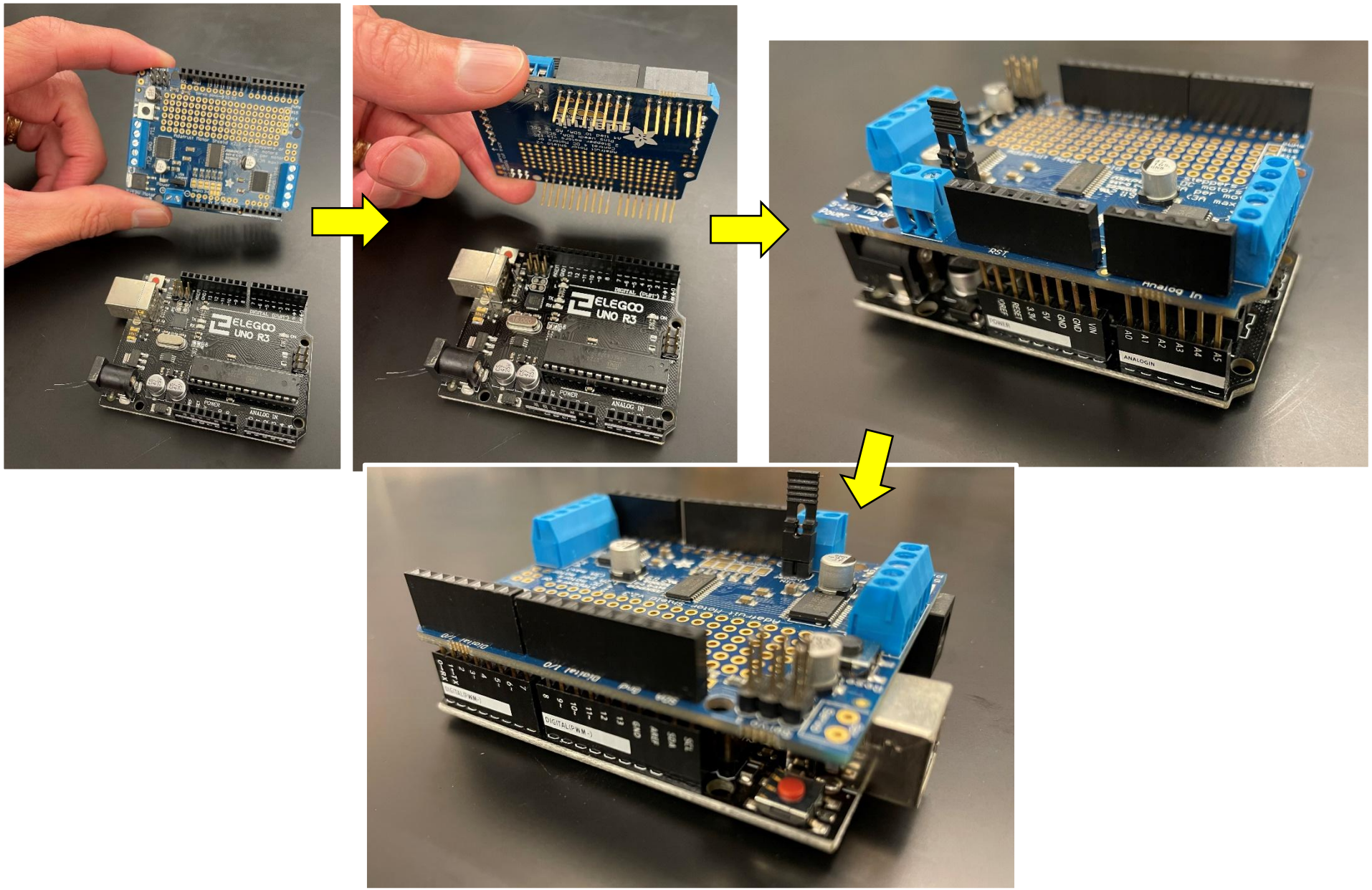
Assembling a Robot Car (cont.)



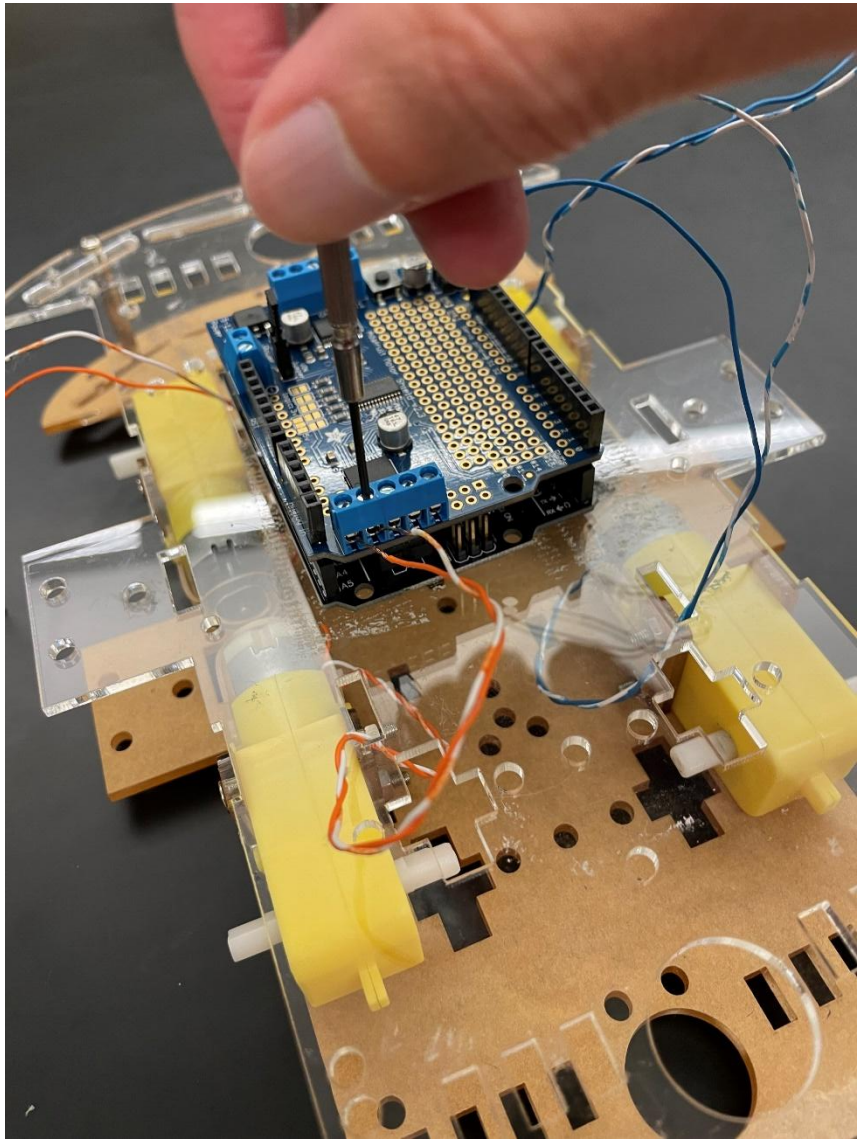


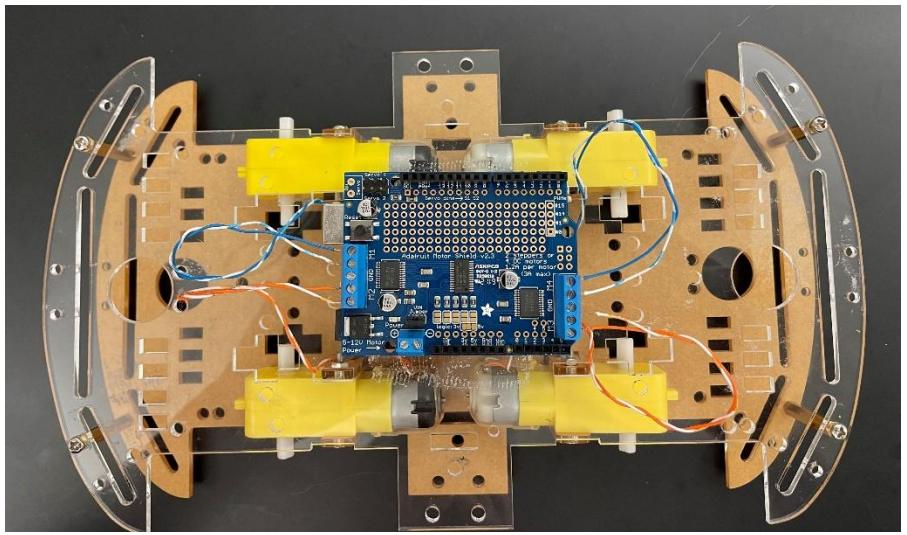


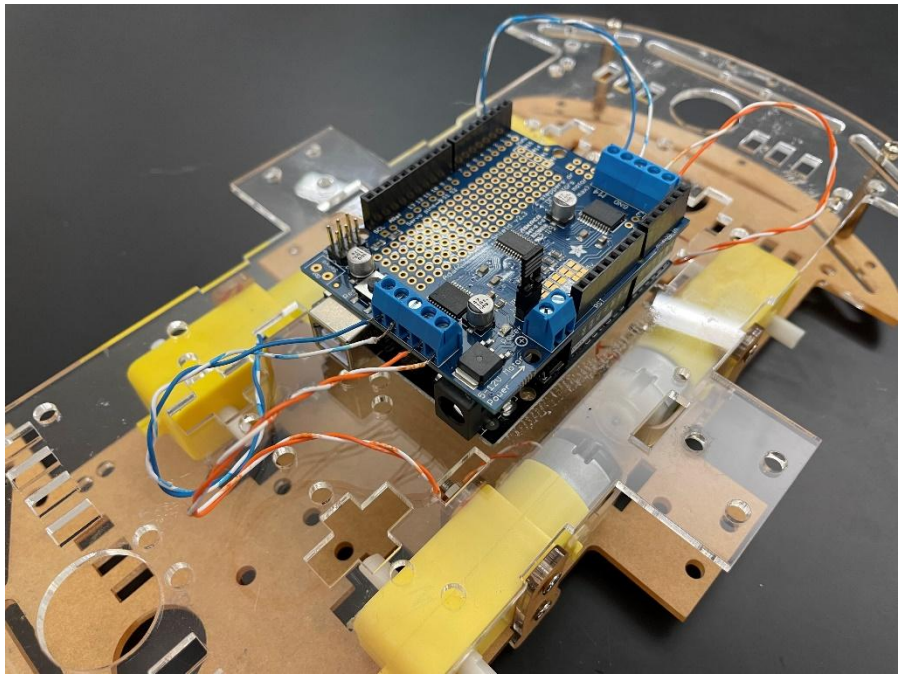
Assembling a Robot Car (cont.)



Assembling a Robot Car (cont.)

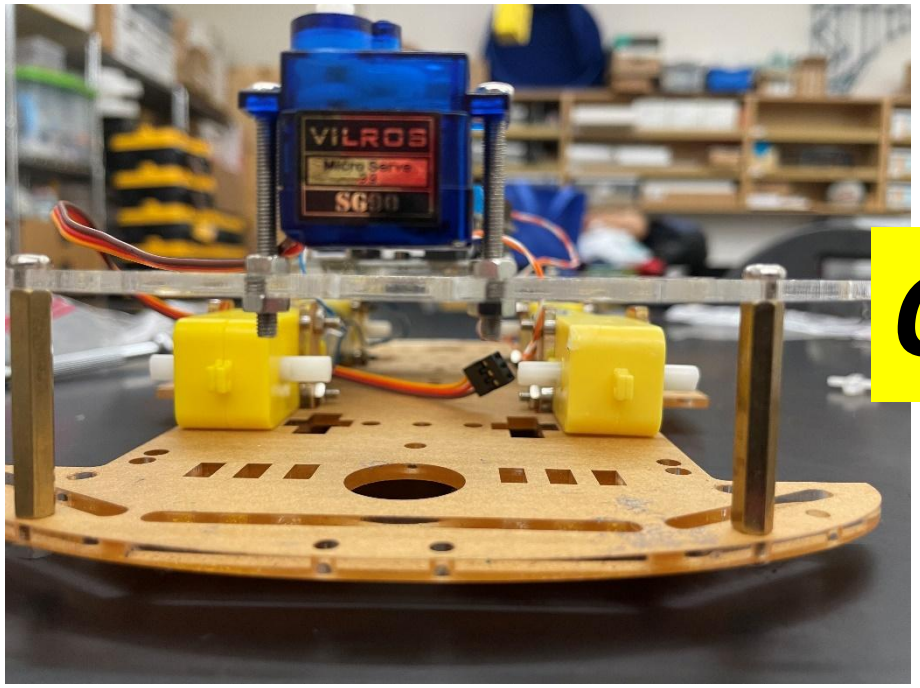






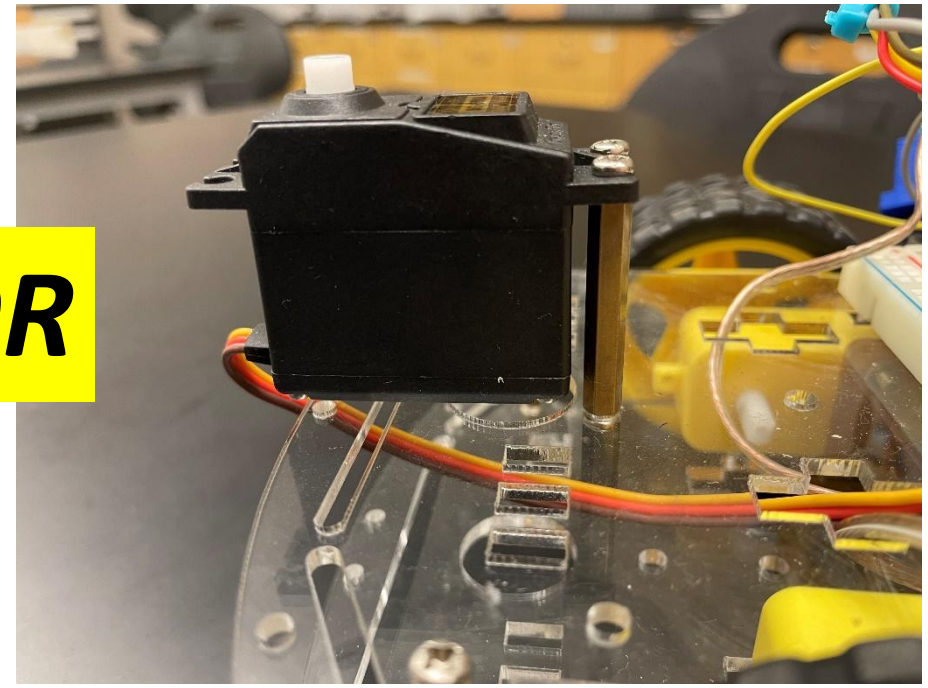
Select a Servo Motor

Small blue servo motor

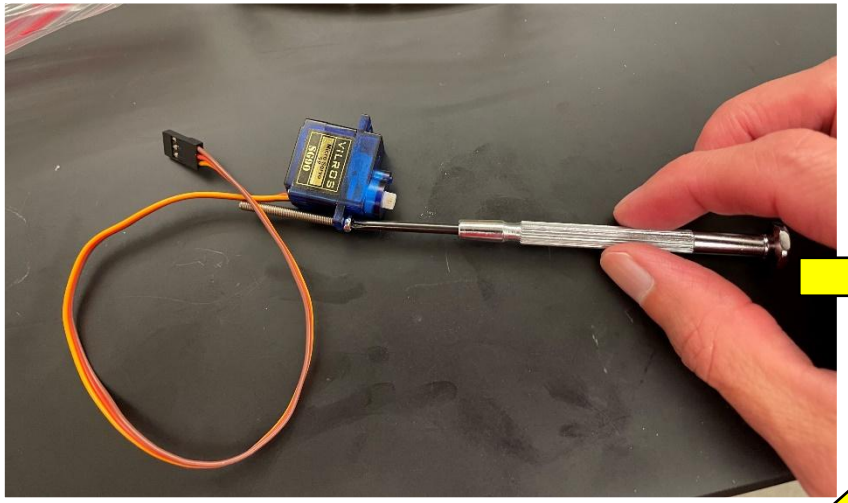


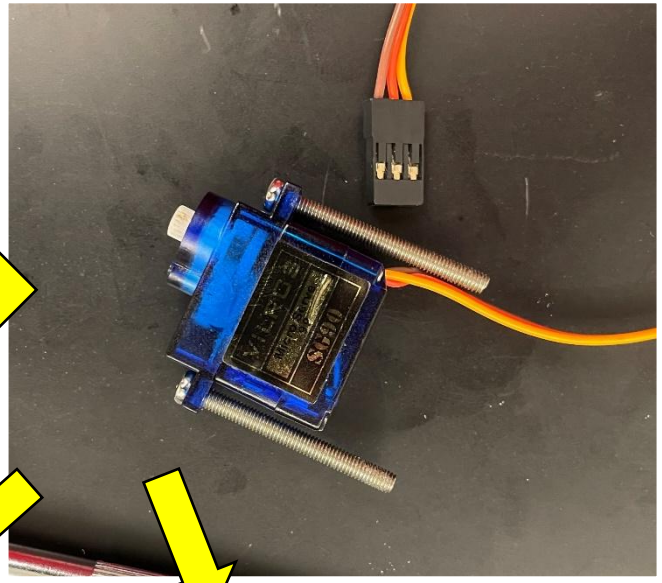
OR

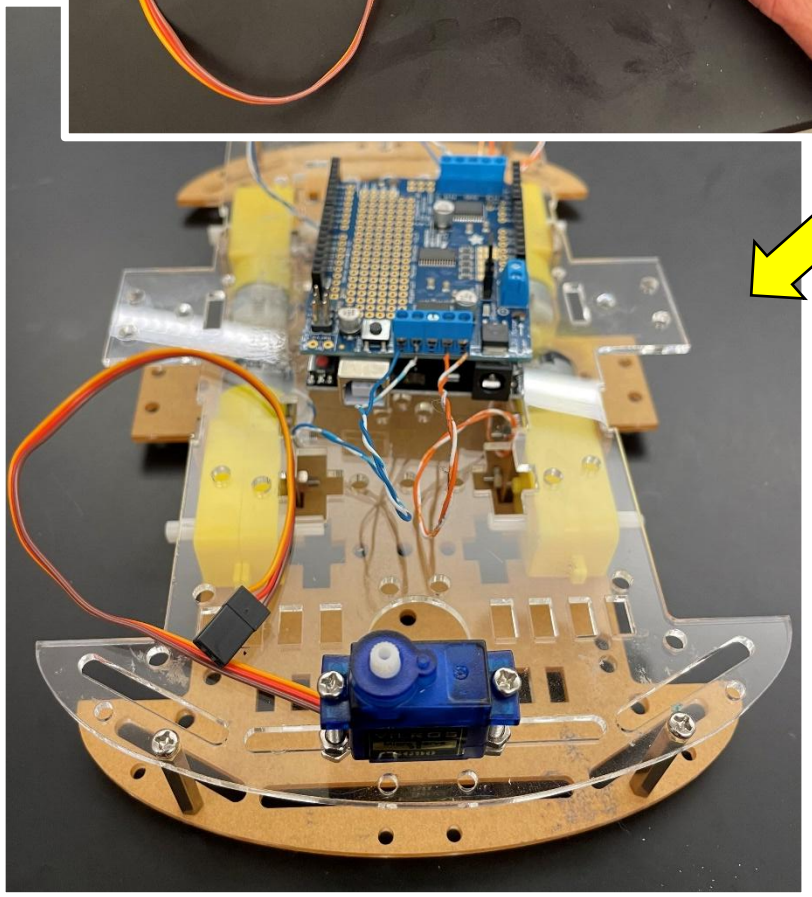
Larger black servo motor

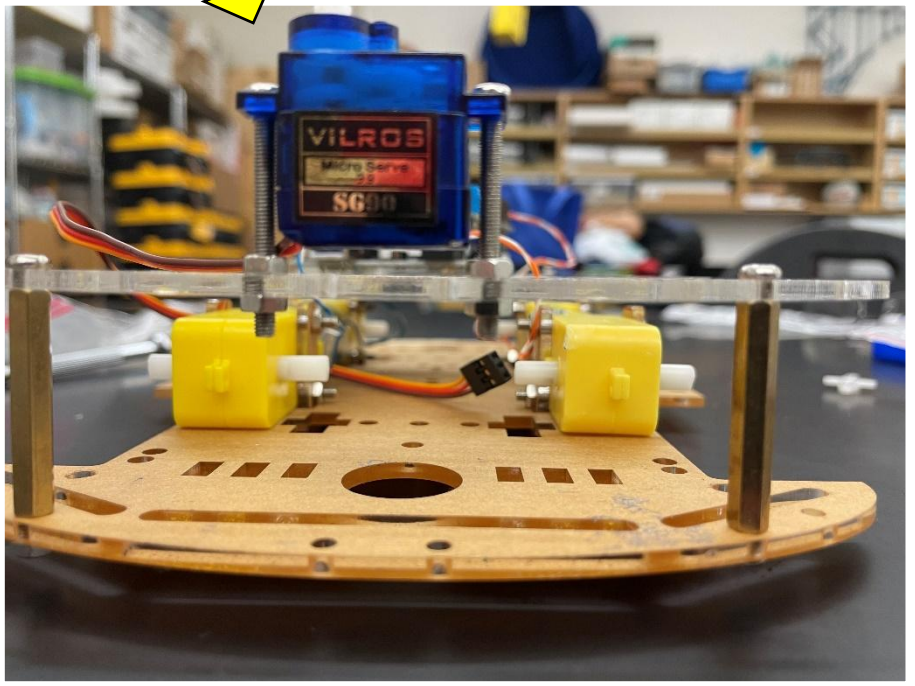


Connecting a Small Blue Servo



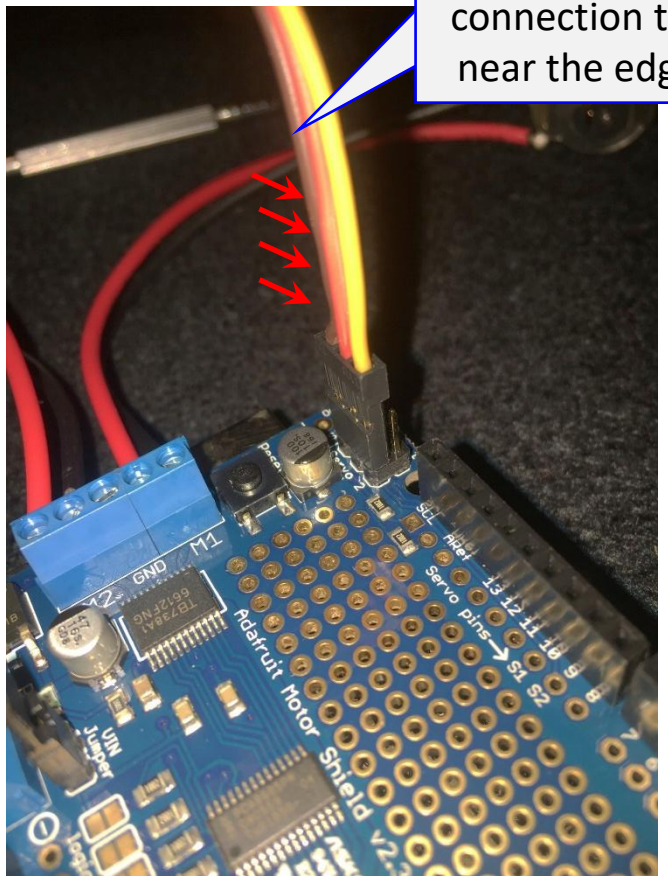


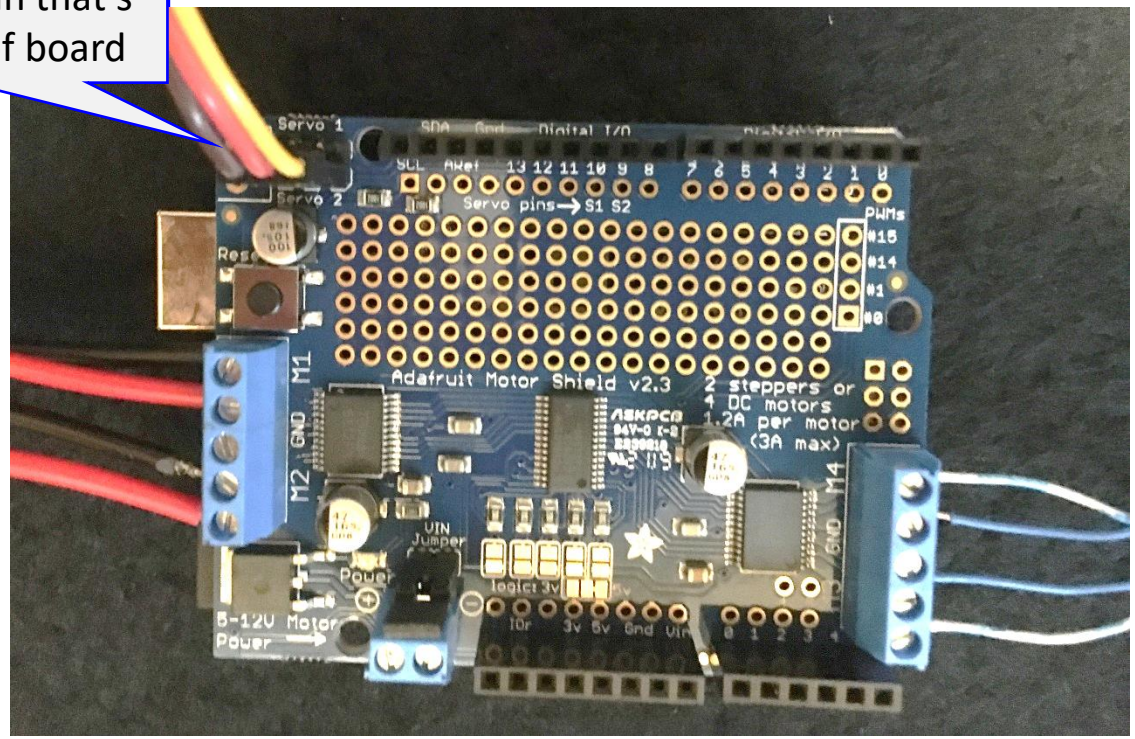




Wiring a Servo Motor

Important:
Connect servo's GND connection to pin that's near the edge of board





Wiring an Ultrasonic Sensor

- Connect the VCC pin from the sensor to 5V on the motor-shield board
- Connect the GND pin from the sensor to GND on the motor-shield
- Connect ECHO and TRIG pins to pins on the motor-shield board that you define in your sketch

